Grades will be notified at

7:00pm of Thursday, June 9, 2022

Question 1 (7.5 points)

In the closed economy Alfa, there are only one-year and two-year bonds. The current and the future expected inflation rates are both zero, so that the real and the nominal interest rates are equal. In period $t+1$ ("the future"), a standard IS-LM model describes the functioning of the economy. In particular, in the year $t+1$ consumption depends only on disposable income at $t+1$, and investment on the interest rate and output at $t+1$ only. Turning now to the current period (t, "the present"), consumption is increasing in both current and expected future disposable incomes, $C_t = C(Y_t - \bar{T}_t, Y^e_{t+1} - \bar{T}^e_{t+1})$, while investment depends positively on current and expected future income levels, and negatively on the current and expected future levels of short-term interest rates. Finally, in both periods the central bank chooses the interest rates and G and T are exogenous as usual.

Assuming that, at time *t*, the current and future expected yields on one-year bonds are $i_{1t} = 2.5\%$ % and $i^e_{1,t+1} = 4.5\%$, write down the numerical values of the two points that Alfa's yield curve connects, and briefly comment on the resulting slope of that curve. Finally, suppose that market participants start expecting that, at time $t+1$, monetary policy will become more restrictive than initially assumed. Discuss if and how these expectations on the future monetary policy stance will affect the slope of time *t* Alfa's yield curve. Explain.

SKETCH OF ANSWER

In this economy where only one-year and two-year bonds exist, the two points that the time t yield curve connects are $i_{1t} = 2.5\%$ *and the yield to maturity of two-year bonds issued at* t, $i_{2t} = \frac{1}{2}\left(i_{1t} + i^e_{1,t+1}\right) = \frac{1}{2}(2.5\% + 4.5\%) = 3.5\%$. *Coherently with the expectations of a rise in short term interest rate that characterize this economy, the yield curve has therefore a positive slope. If at time* $t+1$ *the central bank raises the short term interest rate, the LM curve for that time period shifts upwards and equilibrium income falls. At* t, *the decrease in* Y^e_{t+1} *and the increase in* $i^e_{1,t+1}$ *shift the IS curve to the left. In the new equilibrium,* Y_t *is lower and the current short rate* i_{1t} *unchanged. The yield on two-year bonds will go up by* $\Delta i_{2t} = \frac{1}{2}\left(\Delta i_{1t} + \Delta i^e_{1,t+1}\right) = \frac{1}{2}\left(0 + \Delta i^e_{1,t+1}\right) = \frac{1}{2}\Delta i^e_{1,t+1} > 0$, *implying that the yield curve will remain positively sloped, but will become steeper.*

Question 2 (7.5 points)

Now consider Beta, an economy that differs from Alfa only because, rather than the interest rate, its central bank chooses the money supply. Assuming that, at time *t*, in Beta the current and future expected yields on one-year bonds are the same as in Alfa ($i_{1t} = 2.5\%$ % and $i^e_{1,t+1} = 4.5\%$), repeat the analysis carried out to answer the previous question. In particular, explain how the expectations of a more restrictive monetary policy at time $t + 1$ will affect the slope of the yield curve prevailing in Beta at time t. Explain.

SKETCH OF ANSWER

If the central bank chooses the money supply, the LM curve will be positively sloped in both periods. A restrictive monetary policy implemented at time t+1 will cause a parallel shift to the left of the LM, leading to a higher short term interest rate and a lower equilibrium income in that time period. Both changes will shift the IS curve for time t to the left, thus lowering not just Y_t*, but also* i_{1t}*, the first of the two points connected by the yield curve. As for* i_{2t}*, the time t yield on two-year bonds, it is not possible to say anything definite about the direction in which it will vary: the current short term rate falls, but the future one increases. However, even if* i_{2t} *goes down, being the average of a current short rate that falls and a future expected short rate that rises, it will go down less than* i_{1t}*. It follows that* i_{2t} *will end up being in any case higher than the new* i_{1t}*, so that the new yield curve will be, as concluded when answering the previous question, still positively sloped, and steeper than the initial one.*

Question 3 (7.5 points)

Consider Delta, an economy with constant prices under a fixed exchange rate regime described by a standard open economy "IS-LM-interest parity" model. Delta's central bank is determined to keep the exchange rate constant at the chosen level $E = E_1$, and individuals expect it to do so also in the future, so that $E = E_1 = E^e$. Having explained which value the domestic interest rate will necessarily take on in this economy, suppose that, starting from an equilibrium position, net taxes and autonomous consumption rise in Delta by the same amount, so that $\Delta\bar{T} = \Delta c_0 > 0$. Which of the three curves – IS, LM, interest parity – will be affected by this change, and why? In the move from the initial equilibrium to the new one that the economy will reach, how will Delta's equilibrium levels of production, interest rate, exchange rate, consumption, investment and net exports will change? And what about money supply? Explain.

SKETCH OF ANSWER

- *Under a fixed, and credible, exchange rate ($E = E_1 = E^e$), from the interest parity condition follows that $i = i^*$ – the domestic interest rate must be equal to the foreign one.*
- *The IS curve shifts to the right, since – for any given Y – the simultaneous increase in c_0 and net taxes raises C (to be more precise, C goes up by one minus the marginal propensity to consume times the increase in T – or, which is the same, in c_0). [*increase in *C* "on impact", for given *Y]*
- *The curve that, in the (E, i) plane, represents the interest parity condition is not affected by the change under consideration.*
- *Since, to make sure that the exchange rate remains fixed, the central bank keeps the domestic interest rate equal to the foreign one, the LM curve remains horizontal at $i = i^*$.*
- *In new equilibrium, income is higher (due to the increase in demand that has disturbed the initial equilibrium), i and E unchanged, C higher (both due to its increase on impact, for given Y, mentioned in the second point of this answer, and because of the rise in equilibrium income), I higher (because of the increase Y), NX lower (Y has gone up).*
- *As the economy goes from the initial to the new, final equilibrium, money demand rises (i unchanged; Y higher); it follows that money supply will have to be raised by the same amount money demand has gone up. The central bank will have to increase M to prevent the exchange rate from raising above E_1 – absent this central bank intervention, the increase in money demand would raise the domestic interest rate above the level (i*) it has to take on if the exchange has to remain constant at the chosen level.*

Question 4 (7.5 points)

We are at time t. Assume that the growth rate of the economy is 1% ($g = 0.01$) and that the real interest rate is 6% ($r = 0.06$). Knowing that $B_{t-1}/Y_{t-1} = 1.20$ and $(G_t - T_t)/Y_t = -0.03$, compute the value that the debt ratio will take on at time t. Finally, compute the value of the rate of growth of real GDP, g, that, for the assumed values of the real interest rate (0.06) and the primary deficit-to-GDP ratio (−0.03), would stabilize B/Y at 120% at time t.

From the equation that gives the evolution of the debt ratio $b \equiv B/Y$ over time,

$$b_t = (1 + r - g)b_{t-1} + d_t, \qquad (*)$$

where $d_t \equiv (G_t - T_t)/Y_t$, *it follows that* $b_t = (1 + 0.06 - 0.01) \cdot 1.20 - 0.03 = 1.23$. *Absent any intervention, the debt ratio would therefore increase by* 3 *percentage points. To prevent it from rising, the growth rate of the economy* g *should take on a value leading to* $b_t = b_{t-1} = 1.20$. *Setting* b_t *and* b_{t-1} *equal to* 1.20 *in* $(*)$ *and solving for* g *yields* $g = 0.035$. *[Alternatively, one arrives to the same answer using the definition of steady state debt-to-GDP ratio* $\bar{b} = d/(g - r)$, *setting* $\bar{b} = 1.20$ *and solving the equation* $1.20 = -0.03/(g - 0.06)$ *for* g.*] To stabilize the debt ratio by varying the growth rate of the economy, the government should try to raise* g *to* 3.5%, *from its current level of* 1%.